

PMU Communication Delay and Packet Loss

- Experience from Sweden



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Outline

- Overview of the PMU test network at SvK
- Empirical results
- Identify distributions with best fit
- Possible usages
- Key findings



Main contributors

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 - Sture Holmström
 - Göran Ericsson
- KTH
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 - Davood Babazadeh
 - Lars Nordström



PMU deployment at SvK

- Until Nov 2012, 11 PMUs are in operation.
- *OpenPDC* is implemented to collect synchrophasors for test purposes.
- The configuration of the PMU communication network is compliant with the common practices documented in “Communication Architecture for IP-based Substation Applications”, Cigre D2.28, Report 507, Aug, 2012.



PMU test network in Sweden

- Protocols:
 - Routing protocol: OSPF
 - Transport layer protocol: TCP
- Bandwidth:
 - Substation/Control center LAN: 100 Mbps
 - Core network: 34Mbps
 - Connection between Core and Substation: 2Mbps
- Background traffic:
 - RTU
 - VoIP
 - Video
- Quality of Service-WFQ
 - Priority class: RTU=PMU>VoIP>Video

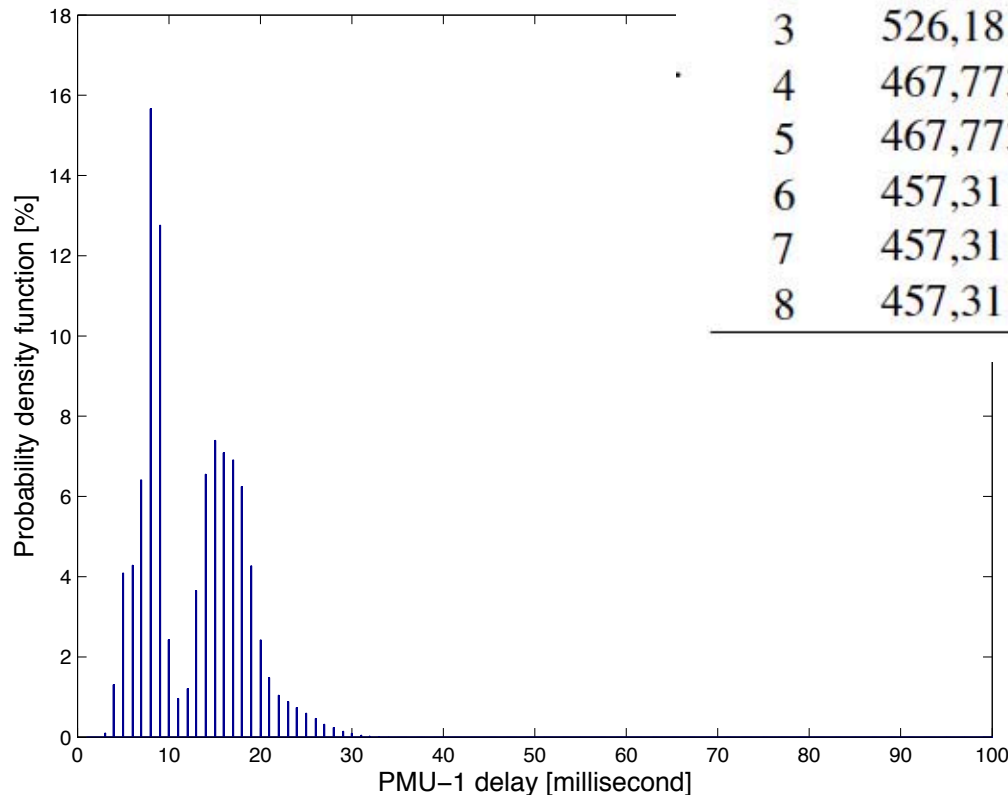
Empirical results

- Average and maximum delay of PMU raw data
 - between PMU and PDC
- Average delay of sorted PMU data
 - between PMU and PDC
 - includes delay due to PDC sorting and downsampling
- Packet loss of PMU raw data
- All the statistics are collected by *OpenPDC* in a time window of 10 seconds between Nov 2012 and Jan 2013.

Average delay of PMU raw data

Table 1: Overview of the raw PMU data stream delay (in milliseconds)

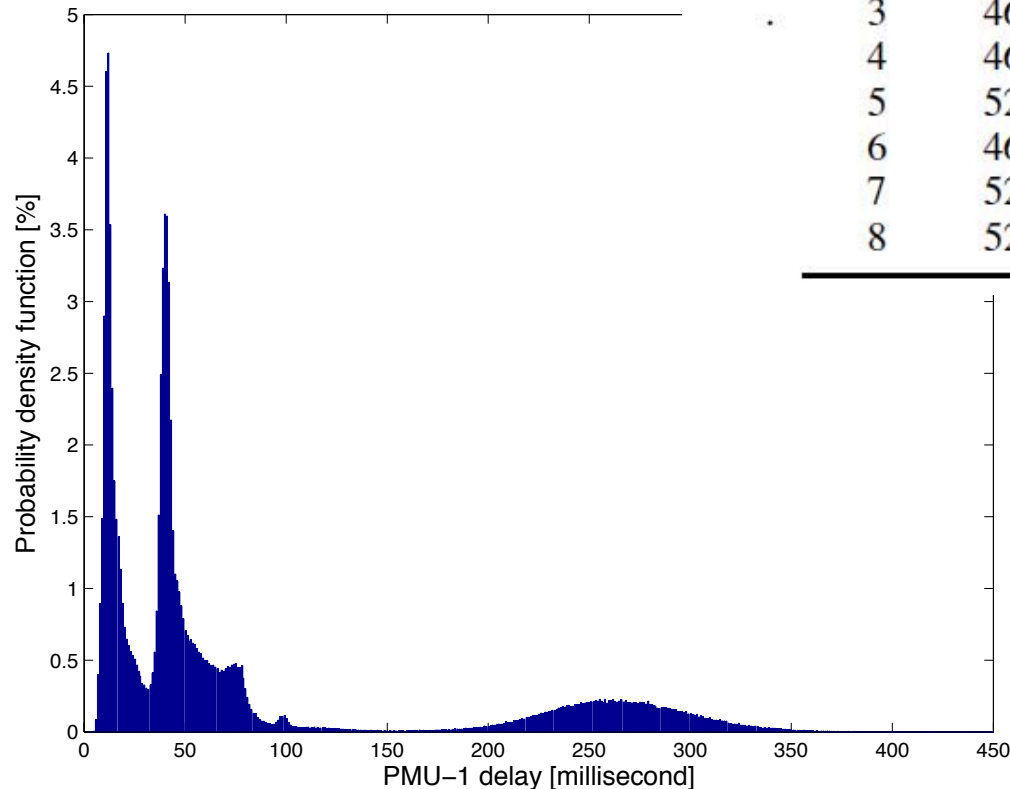
PMU	Samples	Distance	Min/Max	μ/σ	Outliers (>100 ms)
1	467,819	130 km	1/1158	12.66/8.56	0.0618%
2	447,580	480 km	3/3019	9.84/10.48	0.1007%
3	526,181	450 km	1/2390	10.48/10.10	0.0650%
4	467,773	500 km	1/6215	14.13/13.95	0.0767%
5	467,773	500 km	1/5509	14.11/18.53	0.0808%
6	457,311	400 km	8/1073	11.44/6.45	0.1033%
7	457,311	400 km	9/1525	11.54/15.13	0.1122%
8	457,311	400 km	9/1458	11.32/12.67	0.1277%



Maximum delay of PMU raw data

TABLE I: Overview of the maximum delay (in milliseconds) from the studied PMUs

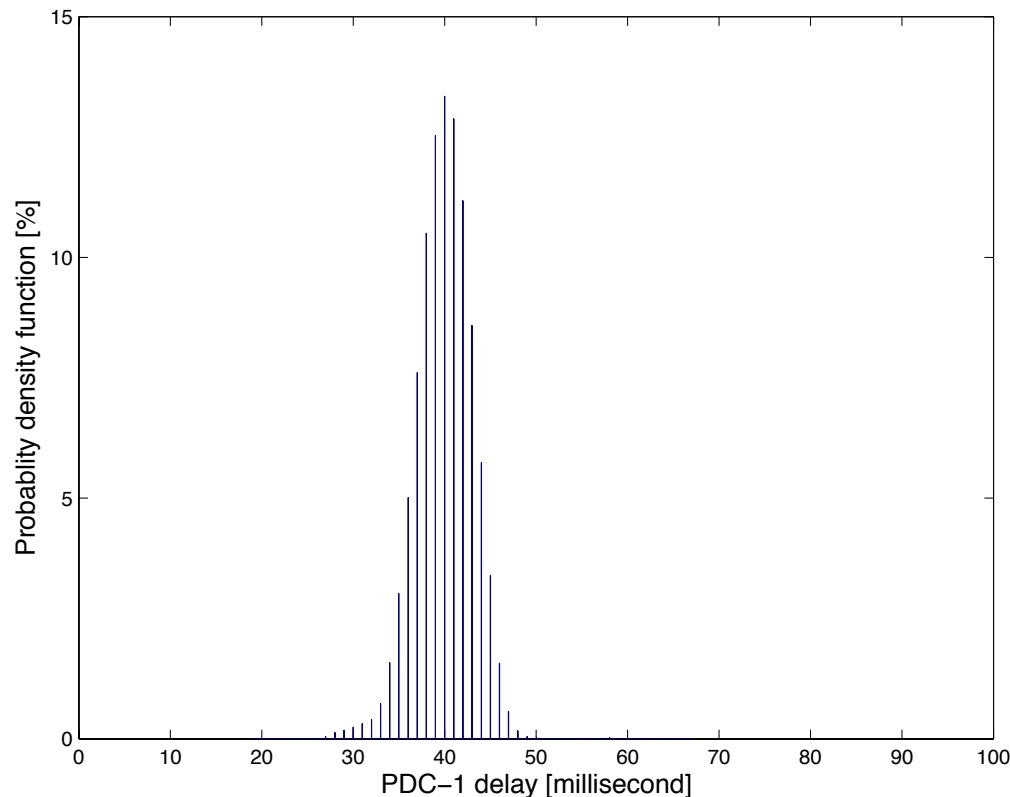
PMU	Samples	Outliers (≥ 450)	Min/Max	μ/σ
1	467,768	0.0725%	2/445	81.05/93.89
2	467,761	0.0688%	1/446	72.25/94.82
3	467,784	0.0724%	1/444	73.62/95.71
4	467,775	0.0894%	4/443	79.09/95.39
5	526,429	0.0914%	3/444	72.27/91.22
6	468,032	0.1030%	2/444	77.54/95.59
7	526,497	0.2089%	2/444	72.32/91.96
8	526,493	0.1274%	2/449	71.68/91.84



Delay of sorted PMU data

Table 2: Overview of the delay (in milliseconds) from sorted synchrophasor streams

PDC	Samples	timeout	Min/Max	μ/σ
1	267,614	23	19/67	40.00/3.11
2	267,614	29	19/67	40.14/3.15



Packet loss rate

- Packet loss rate reported over a long period of time will return optimistic results.

TABLE II: Packet loss rates over a period of two month

PMU	Samples	Anomalies	Packet loss
1	466,345	0.1568%	0.0012%
2	430,649	0.0033%	0.0007%
3	465,528	0.0000%	0.0008%
4	467,087	0.0037%	0.0006%
5	439,373	0.0075%	0.0009%
6	465,534	0.0000%	0.0005%
7	465,528	0.0000%	0.0007%
8	465,529	0.0015%	0.0009%

- The maximum packet loss reached 14% (in a time window of 10 seconds) in our study

Data fitting

- To identify probability distribution offering best fit
- The tested distributions are:
 - Normal
 - Log-normal
 - Generalized Pareto
 - Gamma
 - Weibull
 - Exponential
 - Bi-modal distribution
- Methods
 - Akaike Information Criterion
 - Quantile quantile plot



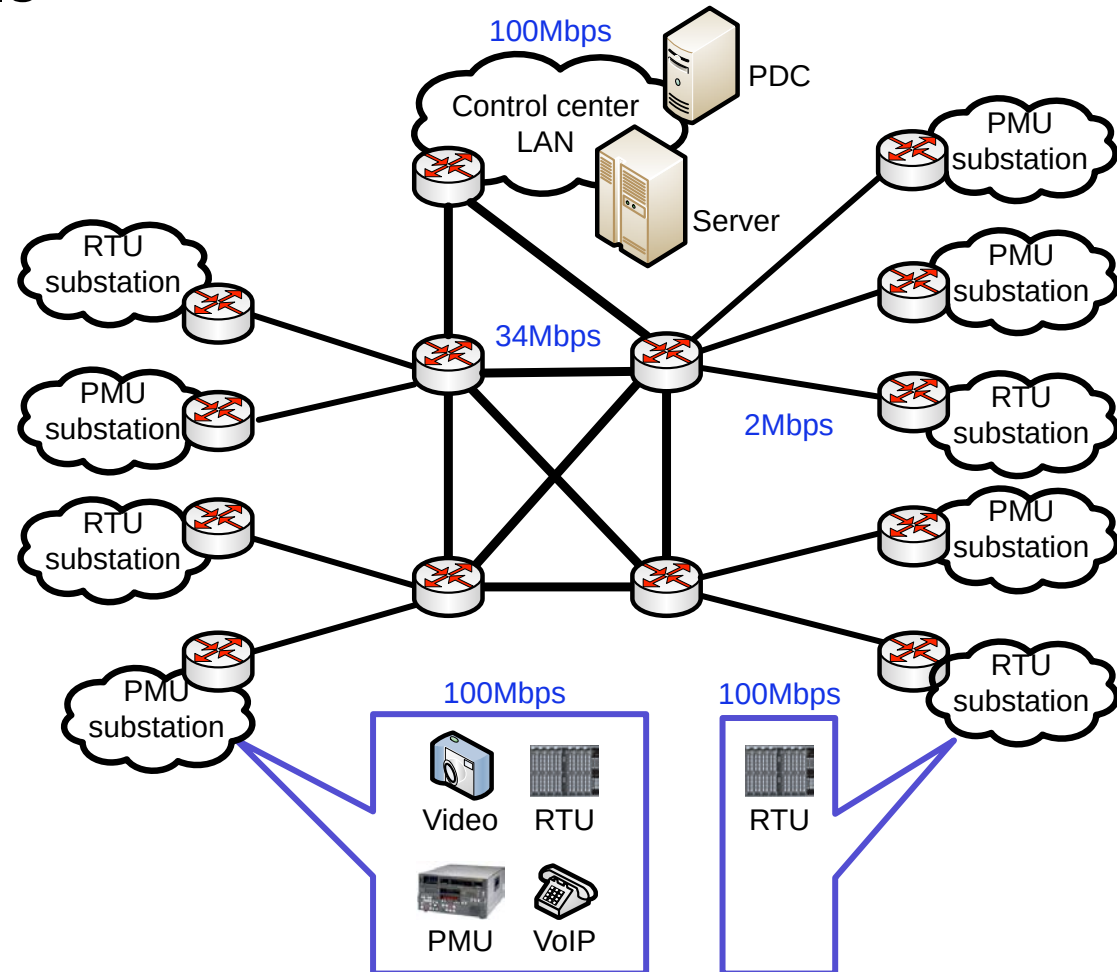
Data fitting

- Bi-modal distribution fits best among all the candidates to model average and maximum delay of raw PMU data.
- Normal distribution provides best fit for the sorted PMU data.

Validation of communication network simulations

- Simulation tools

- NS-2
- OPNET

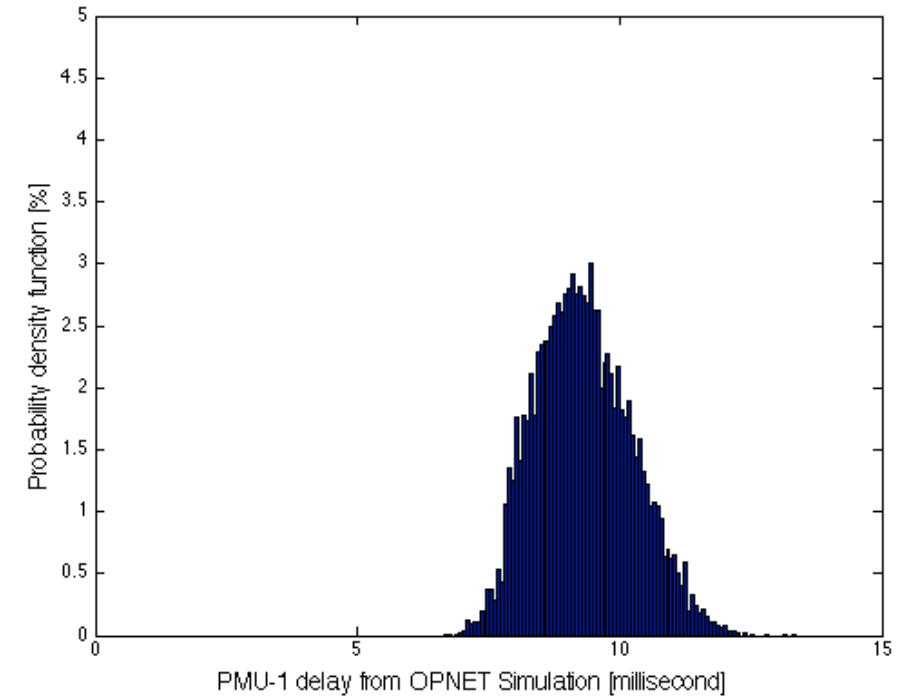
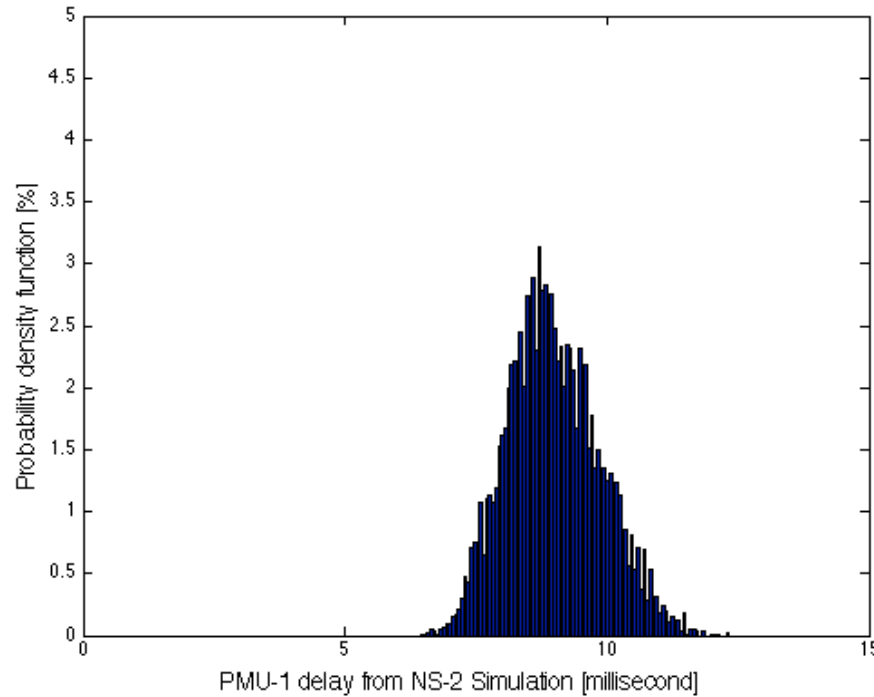




Simulation parameters

- Structure of simulation models
 - perform sensitivity analysis to valid the robustness of the simplification.
- Network parameters (unambiguous)
 - bandwidth
 - protocols
 - Quality of Service
 - Router buffer size
- Traffic profiles (ambiguous)
 - RTU
 - VoIP
 - Video

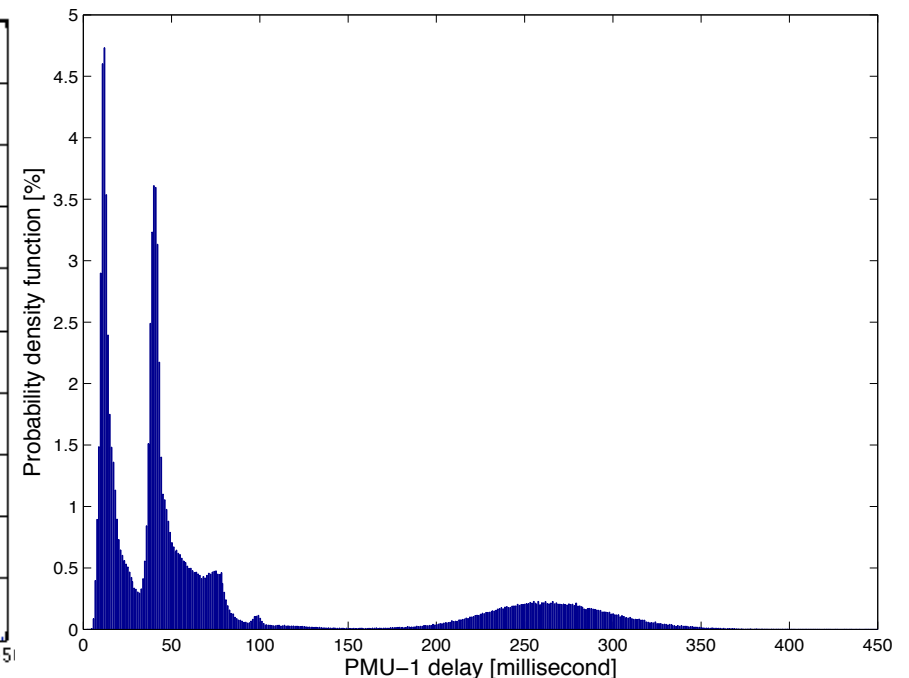
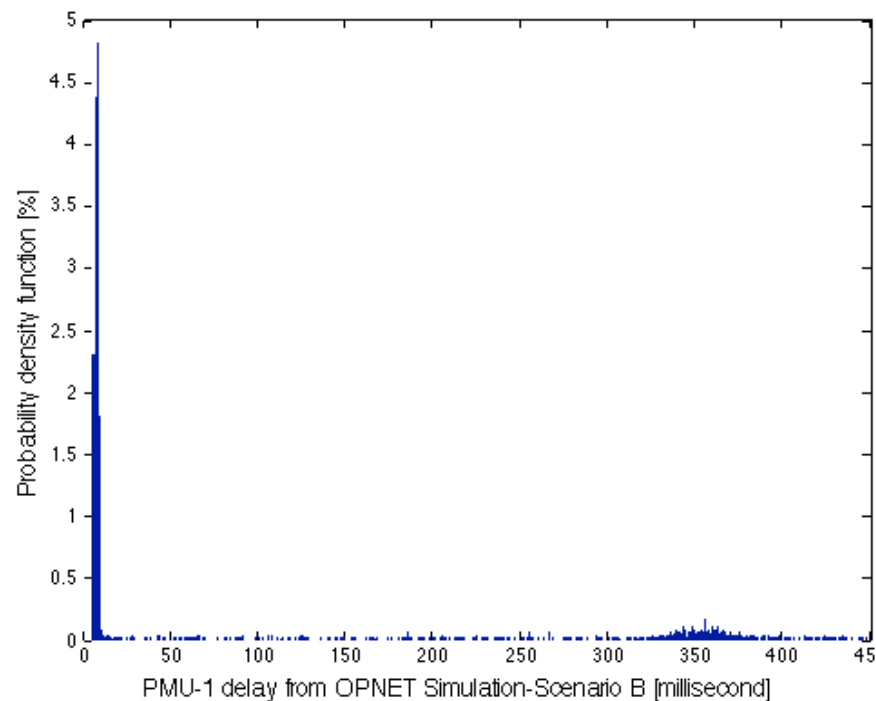
Simulated PMU delay- Scenario A



Type	Packet size (byte)	Reporting rate (packet per second)	Destination
RTU	500	2	Server
PMU	40	50	PDC
VoIP	1024	62	Server
Video	1024	200	Server

Simulated PMU delay- Scenario B

- Scenario B with time-variant RTU traffic profile
 - at maximum the RTU traffic is 5 kbps (base on an empirical study)



Findings

- Simulation of PMU communication may return optimistic delay results
 - unjustified assumptions [1].
 - the communication infrastructure for power system operation and control is a heterogeneous system mixing multiple technologies delivered by different vendors over a long span of time [2].

[1] Barbosa, R.R.R.; Sadre, R.; Pras, A., "A first look into SCADA network traffic," Network Operations and Management Symposium (NOMS), 2012 IEEE, 16-20 April 2012

[2] Wu, F., K. Moslehi, and A. Bose (2005): "Power system control centers: Past,present, and future," Proceedings of the IEEE, 93, 1890 –1908.



Validation of delay robust wide-area damping control schemes

- Augmented delay
 - by applying a liberal assumption that a maximum delay always appears at the first packet available after consecutive packet loss, the augmented delay can be approximated as the sum of maximum delay and delay induced by packet loss.
- Tested delay-robust control schemes
 - Gain scheduling (GSPOD)[3]
 - Generalized predictive control (GPC)[4]
 - Adaptive compensation (APPOD)[5]

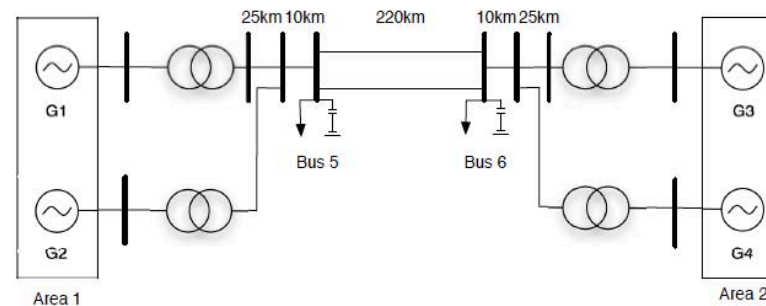


Fig. 4: Four-machine, two-area power system for inter-area oscillation analysis.

- [3] N. Chaudhuri, S. Ray, R. Majumder, and B. Chaudhuri, "A New Approach to Continuous Latency Compensation With Adaptive Phasor Power Oscillation Damping Controller (POD)," *Power Systems, IEEE Transactions on*, vol. 25, no. 2, pp. 939 –946, may 2010.
- [4] H. Wu, K. Tsakalis, and G. Heydt, "Evaluation of time delay effects to wide-area power system stabilizer design," *Power Systems, IEEE Transaction*, vol. 19, no. 4, pp. 1935–1940, Nov 2004.
- [5] W. Yao, L. Jiang, Q. Wu, J. Wen, and S. J. Cheng, "Design of Wide-Area Damping Controllers Based on Networked Predictive Control Considering Communication Delays," in *IEEE PES General Meeting*, Minneapolis, Jul 2010, pp. 1–8.

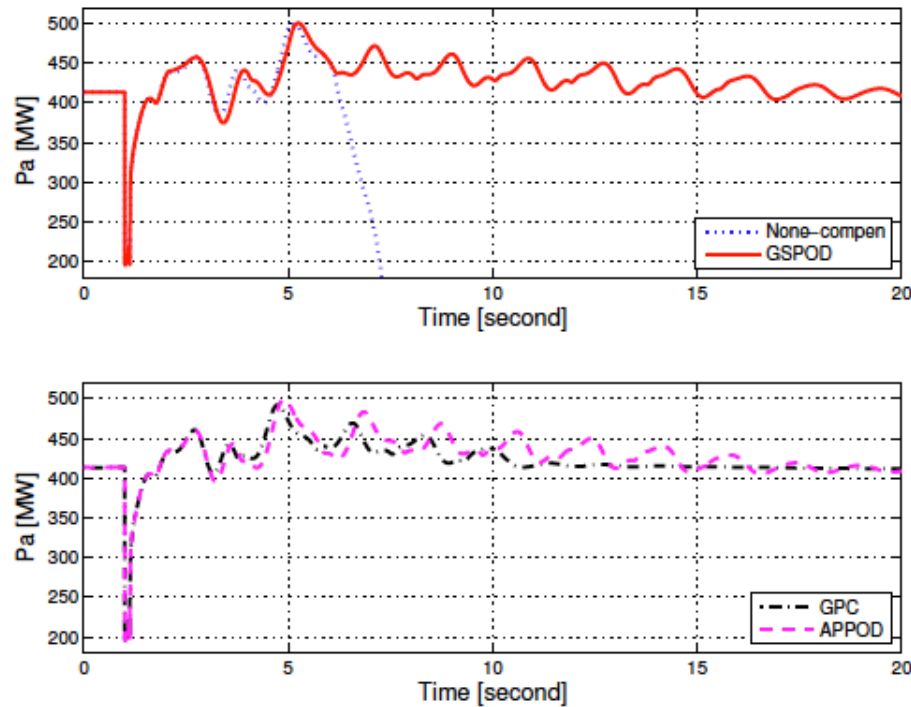


Fig. 5: A snap-shot of system response to three-phase to ground fault, *Scenario A*.

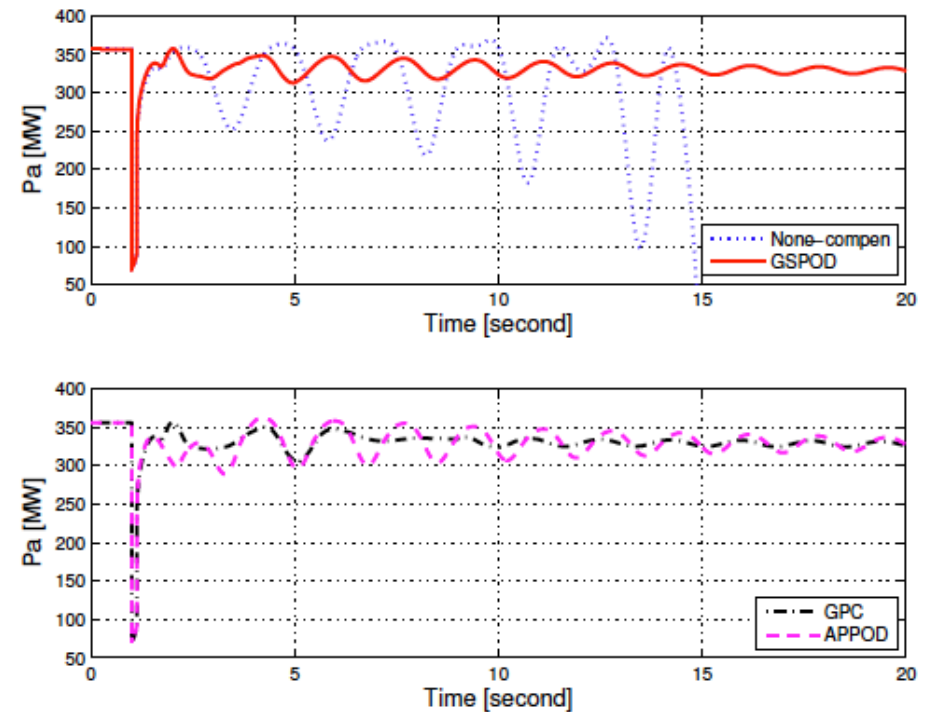


Fig. 6: A snap-shot of system response to three-phase to ground fault, *Scenario B*.

Findings

- Coupled with delay-robust design, this network is capable to provide service to time-stringent synchrophasor based inter-area damping control

TABLE V: Achieved damping control results against the augmented delay profile from *PMU 1*

Scenario	Controller	Overshot $\leq 20\%$	Settling ≤ 15 seconds
A	<i>GSPOD</i>	97.3%	78.3%
	<i>GPC</i>	89.1%	98.3%
	<i>APPOD</i>	92.3%	81.2%
B	<i>GSPOD</i>	96.1%	88.5%
	<i>GPC</i>	82.3%	95.3%
	<i>APPOD</i>	78.8%	92.3%



Details could be accessed at

- K. Zhu, M. Chenine, L. Nordström, Sture Holmström and Göran Ericsson, “An empirical study of synchrophasor delay in a TCP/IP network”, under revision
- K. Zhu, M. Chenine, L. Nordström, Sture Holmström and Göran Ericsson, “ Design of wide-area damping control systems –using empirical results from a utility TCP/IP network”, under review
- K. Zhu, D. Babazadeh, L. Nordström, “Validation of PMU delay simulations with empirical results”, under review